

Research Interests

Transient Radio Astronomy: Fast radio bursts, compact objects, interferometry**Propagation Effects:** Gravitational lensing, multipath scattering, scintillation**Cosmology:** Dark matter, feedback processes, cosmological expansion**Employment**

- 2026-Present **Postdoctoral Researcher**, Laboratoire d'Astrophysique de Marseille, Aix-Marseille University
Fast radio burst Lensing as A Measurement Experiment (FLAME) research group
Supervisor: Dr. Daniele Michilli
- 2025-2026 **FRQNT Fellow**, Trottier Space Institute, McGill University
Supervisors: Prof. Victoria Kaspi, Prof. Matt Dobbs, Prof. Jonathan Sievers
- 2023-2025 **Trottier Fellow**, Trottier Space Institute, McGill University
Supervisors: Prof. Victoria Kaspi, Prof. Matt Dobbs, Prof. Jonathan Sievers

Education

- 2020-2023 **Ph.D. Physics**, Curtin University, Perth, Western Australia
Thesis Title: "*Exploring the Lensing of Cosmological Transients on Nanosecond Timescales*"
Thesis Supervisors: Prof. Cathryn Trott, Dr. Clancy James, Dr. Mark Walker, Prof. Jean-Pierre Macquart.
- 2016-2019 **B.Sc. First Class Honours in Physics** Curtin University, Perth, Western Australia
Thesis Title: "*Fast Radio Burst Temporal Microstructure as a Tool for Hunting Compact Objects*"
Thesis Supervisors: Prof. Jean-Pierre Macquart, Dr. Clancy James
- 2019 **Student Internship**, Pawsey Supercomputing Centre / International Centre for Radio Astronomy Research
- 2018 **Undergraduate Researcher**, Curtin Institute of Radio Astronomy
- 2017 **Student Internship**, Curtin University

Teaching

- 2026 **Co-lecturer**, McGill University, Undergraduate Course - Scientific Computing
• Designed and gave an original lecture and corresponding assignment
- 2025 **Co-lecturer**, McGill University, Graduate Course - High Energy Astrophysics
• Designed and gave an original lecture and corresponding assignment
- 2020-2022 **Laboratory Demonstrator**, Curtin University, Undergraduate Physics
• Updated and led weekly laboratory sessions
• Instructed and graded associated laboratory report writing
• Held office hours to assist with experimental and writing technique.
- 2020 **Grader**, Graduate level general relativity, Curtin University

Supervising

2026	Co-supervisor: Aniket Prasad , Ph.D. Student Project: "<i>FRB Gravitational Lensing with Next Generation Transient Radio Facilities</i>"
2026	Primary Supervisor: Vaishnavi Muralikrishnan , Industry collaborator Project: "<i>Exploring the Latent Space of FRB Scintillation Using Machine Learning</i>"
2026	Primary supervisor: Auden Akinc , Undergraduate Student McGill University Project: "<i>Simulating galaxy cluster strong lensing time delays for FRBs</i>"
2025	Primary supervisor: Airene Ahuja , Undergraduate Student McGill University - Project: "<i>Searching for FRBs traversing Massive Galaxy Clusters</i>" - Result: leading role in article submitted to ApJ (https://arxiv.org/abs/2608.10358)

Leadership Experience

2025-Present	Convener of various working groups within CHIME/FRB and CHORD/FRB
2023-Present	Technical design lead for CHORD/FRB's beamforming and voltage post-processing pipeline

Fellowships, Grants and Awards

Since 2025	FRQNT Fellowship – International, \$90,000, Research
2023-2025	Trottier Fellowship – International, \$60,000, Research
2020-2023	Research Training Program – National, \$90,000, Research
2019	First Class Honours – Institutional, GPA > 85%, Academic
2019	Curtin Honours Scholarship – Institutional, \$5000, Research
2016-2018	Curtin Excellence Scholarship – Institutional, \$15000, Academic
2016-2018	Rotary Tertiary Scholarship – Institutional, \$3750, Academic
2019	Student Internship Stipend , Pawsey Supercomputing Centre/International Centre for Radio Astronomy Research – National, \$6000, Research
2017	Student Internship Stipend , Curtin Institute of Radio Astronomy – National, \$3000, Research

Outreach and Engagement

2025	Invited press interview , CBC Daybreak South, "FRBs A Cosmic Mystery." Host Chris Walker described it as "one of the most clear and engaging science interviews I can recall."
2023-2026	Public Outreach Volunteer Trottier Space Institute - Guided public viewing of the 2024 total solar eclipse - Supported outreach events such as public lectures
2016-2020	Public Observatory Volunteer Perth Observatory - Planned and led weekly night sky tours at Perth Observatory, engaging the public about astronomy while guiding viewing of celestial objects. - Represented the observatory and led public viewing at Astro-Fest, the largest astronomy festival in Western Australia.

Collaboration and Academic Service

2 – 3 per year	Journal Review , Nature Communications, Astrophysical Journal, Physical Review D, Journal of Undergraduate Research in Physics and Astronomy
2026	Scientific Organising Committee SALF XII , Arizona State University
Since 2023	Collaboration Member , CHIME/FRB • Operations Admin: Overseeing telescope operations and troubleshooting technical issues (rotating role with 8 others) • Data Archiving Expert: maintenance and development of the bespoke <code>Datatrail</code> (https://github.com/CHIMEFRB/datatrail-cli) package which tracks and transfers all CHIME data.
Since 2023	Collaboration Member , CHORD • Design and implementation of beamforming and voltage post-processing pipeline: (https://github.com/chord-observatory) • Forecasting FRB rates and planning for novel detection strategies
2024-2025	Summer Undergraduate Selection Committee , Trottier Space Institute
2024-2025	Seminar Committee Member , Trottier Space Institute
2024	Plaskett and Vibert-Douglas Nominations Selection Committee , McGill University
2017-2023	Collaboration Member , CRAFT • Developed fast radio burst morphology fitting routines as part of high time resolution analysis pipeline which operated on raw voltages. • Characterised FRB search sensitivity as a function of burst properties
2020-2022	Development Committee , Curtin Institute of Radio Astronomy • Completed level 1 ALLY training course by Curtin University on 2SLGBTQIA+ awareness and inclusion. • Assisted in organizing institutional celebrations for a variety of cultural holidays • Solicited anonymous feedback on institutional practices and addressed reports of workplace misconduct • Represented the student body on the advisory committee to department executives. • Acted as liaison between the diversity, equity and inclusion branches at each node of the International Centre for Radio Astronomy Research.

Publication List

Summary: 41 papers, 34 refereed publications ; 1,206 citations (1199 on refereed publications); h-index 17; i10-index 23

First Author Publications

08/2026 0 cites	M. W. Sammons , A. Ahuja, M. Dobbs, Z. Kader, E. Davies-Velie, M. Bhardwaj, et al.. “Evidence for Enhancement in the Rate of Fast Radio Bursts Toward Galaxy Clusters.” <i>arXiv e-prints</i> , arXiv:2608.10358 (2026); ADS; doi:10.48550/arXiv.2608.10358.
07/2025 2 cites	M. W. Sammons , E. Davies-Velie, M. Dobbs, Z. Kader, S. R. Siegel, J. Sievers. “Forecasting the Fast Radio Burst Population Observed through Galaxy Cluster Lenses.” <i>The Astrophysical Journal</i> , 987, 139 (2025); ADS; doi:10.3847/1538-4357/addc71.
11/2023 29 cites	M. W. Sammons , A. T. Deller, M. Glowacki, K. Gourdji, C. W. James, J. X. Prochaska, et al.. “Two-screen scattering in CRAFT FRBs.” <i>Monthly Notices of the Royal Astronomical Society</i> , 525, 5653 (2023); ADS; doi:10.1093/mnras/stad2631.

- 12/2022
4 cites **M. W. Sammons**, C. W. James, C. M. Trott, M. Walker. “The effect of gravitational lensing on fast transient event rates.” *Monthly Notices of the Royal Astronomical Society*, 517, 5216 (2022); ADS; doi:10.1093/mnras/stac3013.
- 09/2020
38 cites **M. W. Sammons**, J. P. Macquart, R. D. Ekers, R. M. Shannon, H. Cho, J. X. Prochaska, et al.. “First Constraints on Compact Dark Matter from Fast Radio Burst Microstructure.” *The Astrophysical Journal*, 900, 122 (2020); ADS; doi:10.3847/1538-4357/aba7bb.

Publications with Significant Contributions

- 10/2026
0 cites V. Shah, J. W. T. Hessels, V. M. Kaspi, K. W. Masui, **M. W. Sammons**, D. Amouyal, et al.. “Probing the Maximum Energy of Fast Radio Bursts Using Thousands of Sources from the Second CHIME/FRB Catalog.” *The Astrophysical Journal*, 1009, 155 (2026); ADS; doi:10.3847/1538-4357/ae9a96.
- 10/2025
17 cites D. R. Scott, T. Dial, A. Bera, . . . , **M. Sammons**, et al.. “High-time-resolution properties of 35 fast radio bursts detected by the Commensal Real-time ASKAP Fast Transients survey.” *Publications of the Astronomical Society of Australia*, 42, e133 (2025); ADS; doi:10.1017/pasa.2025.10103.
- 08/2025
27 cites CHIME/FRB Collaboration, T. C. Abbott, D. Amouyal, . . . , **M. W. Sammons**, et al.. “FRB 20250316A: A Brilliant and Nearby One-off Fast Radio Burst Localized to 13 pc Precision.” *The Astrophysical Journal Letters*, 989, L48 (2025); ADS; doi:10.3847/2041-8213/adf62f.
- 12/2024
8 cites Z. Kader, M. Dobbs, C. Leung, K. W. Masui, **M. W. Sammons**. “Simulating FRB morphologies and coherent phase correlation signatures from multiplane astrophysical lensing.” *Physical Review D*, 110, 123027 (2024); ADS; doi:10.1103/PhysRevD.110.123027.
- 09/2024
15 cites A. R. H. Stevens, M. Sinha, A. Rohl, **M. W. Sammons**, B. Hadzhiyska, C. Hernández-Aguayo, et al.. “DARK SAGE: Next-generation semi-analytic galaxy evolution with multidimensional structure and minimal free parameters.” *Publications of the Astronomical Society of Australia*, 41, e053 (2024); ADS; doi:10.1017/pasa.2024.14.
- 10/2023
22 cites L. Marnoch, S. D. Ryder, C. W. James, A. C. Gordon, **M. W. Sammons**, J. X. Prochaska, et al.. “The unseen host galaxy and high dispersion measure of a precisely localized fast radio burst suggests a high-redshift origin.” *Monthly Notices of the Royal Astronomical Society*, 525, 994 (2023); ADS;
- 08/2023
9 cites G. E. Anderson, T. D. Russell, H. M. Fausey, . . . , **M. W. Sammons**, et al.. “Rapid radio brightening of GRB 210702A.” *Monthly Notices of the Royal Astronomical Society*, 523, 4992 (2023); ADS; doi:10.1093/mnras/stad1635.
- 03/2021
37 cites W. R. Arcus, J. P. Macquart, **M. W. Sammons**, C. W. James, R. D. Ekers. “The fast radio burst dispersion measure distribution.” *Monthly Notices of the Royal Astronomical Society*, 501, 5319 (2021); ADS; doi:10.1093/mnras/staa3948.

Contributing Author Publications

- 10/2026
0 cites S. Simha, H. W. Chen, W. f. Fong, . . . , **M. W. Sammons**, et al.. “A spatially resolved view of Fast Radio Burst host metallicities with Integral-field Spectroscopy.” *arXiv e-prints*, arXiv:2610.07325 (2026); ADS.
- 09/2026
0 cites Y. Dong, W. Fong, K. Nimmo, . . . , **M. W. Sammons**, et al.. “Probing the Metallicity Dependence of Fast Radio Burst Progenitors with CHIME/FRB Outtrigger Dwarf Host Galaxies.” *arXiv e-prints*, arXiv:2609.28629 (2026); ADS; doi:10.48550/arXiv.2609.28629.

- 07/2026
0 cites The CHORD Collaboration, K. Bandura, L. Belostotski, . . . , **M. Sammons**, et al.. “Overview of the Canadian Hydrogen Observatory and Radio Transient Detector (CHORD) Project.” *arXiv e-prints*, arXiv:2607.09374 (2026); ADS; doi:10.48550/arXiv.2607.09374.
- 06/2026
0 cites Z. Kader, E. Davies-Velie, M. Dobbs, . . . , **M. Sammons**, et al.. “Detection of partial coherence due to multipath propagation for FRB 20220413B with CHIME/FRB.” *Physical Review D*, 113, 123007 (2026); ADS; doi:10.1103/f2nh-hktz.
- 06/2026
1 cites K. McGregor, J. W. T. Hessels, V. M. Kaspi, . . . , **M. W. Sammons**, et al.. “Debiasing the Observed Fast Radio Burst Population with the CHIME/FRB Selection Function.” *arXiv e-prints*, arXiv:2606.26334 (2026); ADS; doi:10.48550/arXiv.2606.26334.
- 05/2026
0 cites A. P. Curtin, M. Gawroński, J. Hessels, . . . , **M. Sammons**, et al.. “The Astrophysics of Fast Radio Bursts.” *Advancing Astrophysics with the SKA I*, 175 (2026); ADS; doi:10.48550/arXiv.2606.27546.
- 05/2026
7 cites A. M. Cook, K. Shin, Z. Pleunis, . . . , **M. W. Sammons**, et al.. “Discovery of 30 Repeating Fast Radio Burst Sources and Uniform Population Statistics of 80 Repeating Sources from CHIME/FRB.” *arXiv e-prints*, arXiv:2605.08410 (2026); ADS; doi:10.48550/arXiv.2605.08410.
- 05/2026
5 cites A. E. Lanman, S. Simha, K. W. Masui, . . . , **M. Sammons**, et al.. “Constraining Gas Mass Fractions in Galaxy Groups and Clusters with the First CHIME/FRB Outrigger.” *The Astrophysical Journal*, 1003, 5 (2026); ADS; doi:10.3847/1538-4357/ae606e.
- 04/2026
11 cites A. Pandhi, K. Nimmo, S. Andrew, . . . , **M. W. Sammons**, et al.. “A Steadily Declining Dispersion Measure for the Repeating Fast Radio Burst FRB 20220529A: Evidence for a Fast Radio Burst Engine Embedded in an Expanding Supernova Remnant.” *The Astrophysical Journal Letters*, 1000, L53 (2026); ADS; doi:10.3847/2041-8213/ae52f8.
- 03/2026
59 cites CHIME/FRB Collaboration, T. Abbott, B. C. Andersen, . . . , **M. W. Sammons**, et al.. “The Second CHIME/FRB Catalog of Fast Radio Bursts.” *The Astrophysical Journal Supplement Series*, 283, 34 (2026); ADS; doi:10.3847/1538-4365/ae3828.
- 02/2026
19 cites K. Shin, A. Curtin, M. Fine, . . . , **M. Sammons**, et al.. “CHIME/FRB Discovery of the Extremely Active Fast Radio Burst Source FRB 20240114A.” *The Astrophysical Journal*, 997, 334 (2026); ADS; doi:10.3847/1538-4357/ae2fc4.
- 02/2026
7 cites A. P. Curtin, S. Andrew, S. Simha, . . . , **M. W. Sammons**, et al.. “CHIME/FRB Discovery and Localization of the Swift-observed FRB 20241228A.” *The Astrophysical Journal*, 998, 97 (2026); ADS; doi:10.3847/1538-4357/ae2ea0.
- 01/2026
6 cites S. S. Patil, R. A. Main, E. Fonseca, . . . , **M. W. Sammons**, et al.. “A Spatial Gap in the Sky Distribution of Fast Radio Burst Detections Coinciding with Galactic Plasma Overdensities.” *The Astrophysical Journal Letters*, 997, L5 (2026); ADS; doi:10.3847/2041-8213/ae2eb3.
- 11/2025
26 cites CHIME/FRB Collaboration, M. Amiri, B. C. Andersen, . . . , **M. W. Sammons**, et al.. “CHIME/FRB Outriggers: Design Overview.” *The Astrophysical Journal*, 993, 55 (2025); ADS; doi:10.3847/1538-4357/adfdcc.
- 11/2025
14 cites K. Shin, C. Leung, S. Simha, . . . , **M. W. Sammons**, et al.. “Investigating the Sightline of a Highly Scattered Fast Radio Burst through a Cosmic Sheet Structure in the Local Universe.” *The Astrophysical Journal*, 993, 208 (2025); ADS; doi:10.3847/1538-4357/ae093b.
- 10/2025
7 cites Y. Dong, C. D. Kilpatrick, W. Fong, . . . , **M. Sammons**, et al.. “Searching for Historical Extragalactic Optical Transients Associated with Fast Radio Bursts.” *The Astrophysical Journal*, 991, 199 (2025); ADS;

doi:10.3847/1538-4357/adfb74.

- 10/2025
27 cites A. P. Curtin, K. R. Sand, Z. Pleunis, . . . , **M. W. Sammons**, et al.. “Morphology of 35 Repeating Fast Radio Burst Sources at Microsecond Time Scales with CHIME/FRB.” *The Astrophysical Journal*, 992, 206 (2025); ADS; doi:10.3847/1538-4357/adf844.
- 09/2025
16 cites C. Leung, S. Simha, I. Medlock, . . . , **M. W. Sammons**, et al.. “Stellar Mass-Dispersion Measure Correlations Constrain Baryonic Feedback in Fast Radio Burst Host Galaxies.” *The Astrophysical Journal Letters*, 991, L25 (2025); ADS; doi:10.3847/2041-8213/ae044d.
- 09/2025
61 cites CHIME/FRB Collaboration, M. Amiri, D. Amouyal, . . . , **M. W. Sammons**, et al.. “A Catalog of Local Universe Fast Radio Bursts from CHIME/FRB and the KKO.” *The Astrophysical Journal Supplement Series*, 280, 6 (2025); ADS; doi:10.3847/1538-4365/addbda.
- 08/2025
3 cites M. Bhardwaj, V. M. Kaspi, K. W. Masui, B. M. Gaensler, A. L. Ibik, **M. W. Sammons**. “Constraining the Faint-end Slope of the Fast Radio Burst Energy Function Using CHIME/FRB Catalog-1 and Local Volume Galaxies.” *The Astrophysical Journal*, 989, 130 (2025); ADS; doi:10.3847/1538-4357/aded17.
- 02/2025
30 cites K. R. Sand, A. P. Curtin, D. Michilli, . . . , **M. W. Sammons**, et al.. “Morphology of 137 Fast Radio Bursts Down to Microsecond Timescales from the First CHIME/FRB Baseband Catalog.” *The Astrophysical Journal*, 979, 160 (2025); ADS; doi:10.3847/1538-4357/ad9b11.
- 02/2025
48 cites T. Eftekhari, Y. Dong, W. Fong, . . . , **M. Sammons**, et al.. “The Massive and Quiescent Elliptical Host Galaxy of the Repeating Fast Radio Burst FRB 20240209A.” *The Astrophysical Journal Letters*, 979, L22 (2025); ADS; doi:10.3847/2041-8213/ad9de2.
- 02/2025
56 cites V. Shah, K. Shin, C. Leung, . . . , **M. Sammons**, et al.. “A Repeating Fast Radio Burst Source in the Outskirts of a Quiescent Galaxy.” *The Astrophysical Journal Letters*, 979, L21 (2025); ADS; doi:10.3847/2041-8213/ad9ddc.
- 01/2025
67 cites K. Nimmo, Z. Pleunis, P. Beniamini, . . . , **M. W. Sammons**, et al.. “Magnetospheric origin of a fast radio burst constrained using scintillation.” *Nature*, 637, 48 (2025); ADS; doi:10.1038/s41586-024-08297-w.
- 10/2024
14 cites A. M. Cook, P. Scholz, A. B. Pearlman, . . . , **M. W. Sammons**, et al.. “Contemporaneous X-Ray Observations of 30 Bright Radio Bursts from the Prolific Fast Radio Burst Source FRB 20220912A.” *The Astrophysical Journal*, 974, 170 (2024); ADS; doi:10.3847/1538-4357/ad6a13.
- 10/2023
104 cites S. D. Ryder, K. W. Bannister, S. Bhandari, . . . , **M. W. Sammons**, et al.. “A luminous fast radio burst that probes the Universe at redshift 1.” *Science*, 382, 294 (2023); ADS; doi:10.1126/science.adf2678.
- 03/2020
122 cites H. Cho, J. P. Macquart, R. M. Shannon, . . . , **M. W. Sammons**, et al.. “Spectropolarimetric Analysis of FRB 181112 at Microsecond Resolution: Implications for Fast Radio Burst Emission Mechanism.” *The Astrophysical Journal Letters*, 891, L38 (2020); ADS; doi:10.3847/2041-8213/ab7824.
- 10/2018
288 cites R. M. Shannon, J. P. Macquart, K. W. Bannister, . . . , **M. Sammons**, et al.. “The dispersion-brightness relation for fast radio bursts from a wide-field survey.” *Nature*, 562, 386 (2018); ADS; doi:10.1038/s41586-018-0588-y.

Conferences and Seminars

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- Sep 2026 **Invited Seminar**, Dominion Radio Astrophysical Observatory Science Seminar DRAO, Penticton, Canada. “Evidence for Enhancement in the Rate of Fast Radio Bursts Toward Galaxy Clusters.” (25+5 minutes)
- Aug 2026 **Contributed talk**, Dynamic Radio Sky 2026 International Conference, Montreal, Canada. “Evi-

- dence for Enhancement in the Rate of Fast Radio Bursts Toward Galaxy Clusters.” (12+3 minutes)
- Jun 2026 **Contributed talk**, FRB 2026 International Conference, Guiyang, China. “Evidence for Enhancement in the Rate of Fast Radio Bursts Toward Galaxy Clusters.” (12+3 minutes)
- Dec 2025 **Invited talk**, 33rd Texas Symposium on Relativistic Astrophysics, Arizona, United States. “A New Era of Fast Radio Burst Science with the Canadian Hydrogen Observatory and Radio Transient Detector.” (12+3 minutes)
- Jul 2025 **Contributed talk**, Dynamic Radio Sky 2025, Sydney, Australia. “How to Search for High Redshift FRBs Using Strong Lensing Galaxy Clusters” (12+3 minutes)
- Jul 2025 **Contributed talk**, FRB 2025 International Conference, Montreal, Canada. “FRB 20250316A: A Brilliant and Nearby One-Off Fast Radio Burst Localized to 13 parsec Precision.” (12+3 minutes; recording)
- Jul 2025 **Contributed talk**, Astronomical Society of Australia Annual Science Meeting 2025, Online. “Finding High Redshift FRBs with Galaxy Clusters.” (12+3 minutes; recording)
- Nov 2023 **Contributed talk**, FRB 2023 International Conference, Bhopal, India. “Two-Screen Scattering in CRAFT FRBs.” (12+3 minutes)
- Nov 2023 **Contributed talk**, Scintillometry 2023 International Conference, Taipei, Taiwan. “Two-Screen Scattering in CRAFT FRBs.” (15+5 minutes; slides)
- Jun 2023 **3rd Ph.D. milestone seminar**, Curtin University. “Exploring the Lensing of Cosmological Transients on Nano-second timescales.” (30+10 minutes)
- Nov 2022 **Invited seminar**, Max-Planck Institute for Radio Astronomy, Bonn, Germany. “Fast Radio Burst Gravitational Lensing and Scintillation.” (30+10 minutes)
- Nov 2022 **Invited seminar**, University of Amsterdam, Amsterdam, Netherlands. “Fast Radio Burst Gravitational Lensing and Scintillation.” (30+10 minutes)
- Nov 2022 **Invited seminar**, Heidelberg University, Heidelberg, Germany. “Fast Radio Burst Gravitational Lensing and Scintillation.” (45+15 minutes)
- Aug 2022 **Contributed talk**, International Astronomical Union General Assembly, Busan, South Korea. “Effect of Gravitational Lensing on Fast Transient Event Rates.” (8+2 minutes; recording)
- Jul 2022 **Contributed talk**, Astronomical Society of Australia Annual Science Meeting 2021, Online. “Effect of Gravitational Lensing on Fast Transient Event Rates.” (8+2 minutes)
- Sep 2021 **2nd Ph.D. milestone seminar**, Curtin University. “Exploring the Lensing of Cosmological Transients on Nano-second timescales.” (45+15 minutes)
- Jul 2021 **Contributed talk**, Astronomical Society of Australia Annual Science Meeting 2021, Online. “Effect of Gravitational Lensing on Fast Transient Event Rates.” (8+2 minutes)
- Jul 2021 **Contributed talk**, FRB 2021 International Conference, Online. “Effect of Gravitational Lensing on Fast Transient Event Rates.” (8+2 minutes)
- Sep 2020 **1st Ph.D. milestone seminar**, Curtin University. “Exploring the Lensing of Cosmological Transients on Nano-second timescales.” (30+10 minutes)